ODRA BIRA

AI/ML Engineer · LLM Applications & RAG · Generative Models · Computer Vision

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SUMMARY

M.S. Computer Science (AI/ML) candidate at the University at Buffalo (**3.83 GPA**) with **3 engineering roles** in deep-learning research and full-stack development. Ships LLM apps end to end (multi-provider FastAPI, hybrid RAG) and trains diffusion and vision models on GPUs.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C/C++
Generative AI & LLMs: Claude & Gemini APIs, Ollama, vLLM, RAG, embeddings, vector search, BM25 hybrid retrieval, cross-encoder reranking, structured outputs, prompt engineering, LoRA/QLoRA fine-tuning (Hugging Face TRL, PEFT)
Machine Learning & CV: PyTorch, TensorFlow/Keras, scikit-learn, diffusion models, U-Net, YOLOv8, OpenCV, MediaPipe, transfer learning
Backend & Data: FastAPI, Flask, Pydantic, RESTful APIs, React, Node.js, PostgreSQL, MongoDB, Spark, Pandas, NumPy
MLOps & Tools: Git, Docker, Linux, CUDA GPU servers (SSH, tmux), pytest, Weights & Biases, Gradio, Streamlit

EXPERIENCE

University at Buffalo | *Graduate Student Researcher* *Oct 2025 – Present*

- Reproduce and benchmark safety-constrained generative planners (**SafeDiffuser, ICLR 2025**) as baselines for lab research on safe planning with diffusion and flow-matching models.
- Trained a **temporal U-Net diffusion trajectory planner** (600-step horizon, 1M iterations) and a value function for guided sampling on D4RL Walker2d on an **NVIDIA RTX 5090** GPU server.
- Benchmarked QP-based vs. closed-form **control barrier function (CBF)** enforcement over 100 episodes each: both met the safety constraint, and the closed form cut planning latency **3.2× (0.64 s → 0.20 s)**.
- Made the research code reproducible: ported it to Python 3.10 and fixed device, rendering and dependency conflicts.

Accentiqa Systems Pvt. Ltd. | *Full Stack Developer* *Jun 2025 – Aug 2025*

- Delivered full-stack features for enterprise workflow apps, owning both the React front end and the RESTful back-end services.
- Designed relational database schemas and built end-to-end CRUD flows across UI, API and data layers, with validated React forms handling state, errors and client–server validation.

Risposta Software India Pvt. Ltd. | *Software Developer Intern* *May 2024 – Aug 2024*

- Cut API data latency **20%** by profiling the slow SQL queries behind REST endpoints and rewriting them.
- Cut page-load time **30%** by reworking how React/Node.js features fetched and rendered data; built React UI and REST APIs.

AI PROJECTS

StudyEase – LLM-Powered Study Planner | *FastAPI, React, Pydantic, Claude/Gemini, Ollama, pytest* github.com/odra-bira/Studyplanner

- Architected a full-stack LLM app (React/Vite + async FastAPI, ~1.5K-line Python back end) exposing **5 typed REST endpoints** for planning, rebalancing, task breakdown, chat and health, plus PDF/image upload with pypdf extraction.
- Designed a **provider-agnostic LLM layer** (adapter pattern) over Claude, Gemini and local Ollama, switchable by config.
- Built a deterministic **validation-and-repair engine** that enforces date windows, daily capacity and 15–180 min session bounds on every LLM-generated plan, with rule-based fallback on model failure; **35 pytest tests** cover all layers.

SkinGPT – RAG Knowledge Assistant | *Embeddings, vector search, BM25, cross-encoder, LLM APIs*

- Engineered a multi-stage **RAG pipeline** (chunking → embedding index → hybrid BM25 + dense retrieval → cross-encoder reranking) that generates citation-grounded answers over unstructured documents.
- Built a **1,000-question evaluation benchmark** to measure retrieval quality and drive tuning, reaching **~75% retrieval precision**.

ShadowFix – Deep Learning Image Restoration | *PyTorch, U-Net, AMP, OpenCV, Gradio* github.com/odra-bira/ShadowFix

- Trained 3 U-Net variants (RGB, residual, mask-guided; ~7.8M params) on ISTD with mixed precision on an NVIDIA T4. Mask conditioning reached **30.19 dB PSNR / 0.951 SSIM** on 540 test images: **+3.0 dB** over the RGB baseline and +9.6 dB over the raw input.
- Split PSNR by region to show residual models over-correcting non-shadow pixels (−1.3 dB); deployed the best model as a Gradio app with automatic (OpenCV LAB-space mask) and paint-guided modes.

Trash Detection & Cleanliness Scoring | *YOLOv8n, OpenCV, Gradio, pytest* github.com/odra-bira/trash_detection

- Fine-tuned YOLOv8n on TACO (30 epochs, 640 px) to **0.78 precision**, tracing the recall gap to small objects; built a 0–100 cleanliness score and density heatmap, shipped as a CLI + Gradio app with pytest tests.

EDUCATION

University at Buffalo, SUNY · M.S. Computer Science (AI/ML) · GPA 3.83/4.00 Aug 2025 – Present
Coursework: Machine Learning, Deep Learning, Data-Intensive Computing, Statistical Data Mining, Algorithms, Databases, OS

Keshav Memorial Institute of Technology (JNTUH) · B.Tech. Computer Science (AI & ML) · GPA 8.4/10 Aug 2021 – May 2025

PUBLICATION

Bira, O., et al. "Automated Assessment of Fruit Quality Using Transfer Learning and MobileNetV2." *IJRASET* 13(4), 2025.